Examiner only

13. *(a)* Potassium dichromate(VI) is used as a reagent in acid solution in a wide range of redox reactions. The dichromate(VI) ions can be reduced using **excess** metallic zinc in acid conditions.

- Elfed performs the experiment in a sealed tube under an atmosphere of pure nitrogen gas
- Fatima performs the experiment in a standard test tube

The final products are different in each case. Use the standard electrode potentials below to identify the products in both cases. Give reasons for your answers. [4]

	Standard electrode potential, E^{θ} / V
$Cr^{3+}(aq) + 3e^- \rightleftharpoons Cr(s)$	–0.78
$Zn^{2+}(aq) + 2e^- \rightleftharpoons Zn(s)$	–0.76
$Cr^{3+}(aq) + e^- \rightleftharpoons Cr^{2+}(aq)$	–0.42
$O_2(g) + 2H_2O(l) + 4e^- \rightleftharpoons 4OH^-(aq)$	+0.40
$Cr_2O_7^{2-}(aq) + 14H^+(aq) + 6e^- \rightleftharpoons 2Cr^{3+}(aq) + 7H_2O(l)$	+1.33

....

....

....

....

....

....

....

....

Examiner only

(b) Chromium(VI) is present in the anions dichromate(VI), $Cr_2O_7^{2-}$, and chromate(VI), CrO_4^{2-}. These ions can be interconverted in solution with the following equilibrium being established.

$$Cr_2O_7^{2-} + H_2O \rightleftharpoons 2CrO_4^{2-} + 2H^+$$

The equilibrium constant, K_c, for this reaction is $2.42 \times 10^{-15}\,mol^2\,dm^{-6}$.

In general, dichromate(VI) is the main species in acid solution and chromate(VI) is the main species in alkaline solution.

(i) Write an expression for the equilibrium constant, K_c, for this reaction. [1]

(ii) The mass of the water in $1.00\,cm^3$ of a dilute aqueous solution is 1.00 g. Show that the concentration of water in any dilute aqueous solution is $55.5\,mol\,dm^{-3}$. [2]

Examiner only

(iii) A solution was prepared containing equal concentrations of $Cr_2O_7^{2-}$ and CrO_4^{2-}. The concentration of each was 0.057 mol dm^{-3}. A student predicted that a solution containing equal amounts of both must be neutral. Is the student correct? Use calculation(s) to justify your answer. [4]

....

....

....

....

(iv) The value of the equilibrium constant for this reaction increases when the mixture is warmed. State and explain what information this provides about the enthalpy change of the reaction. [2]

....

....

....

....

13

END OF PAPER

Surname	Centre Number	Candidate Number
First name(s)		2

GCE A LEVEL

1410U30-1

Z22-1410U30-1

THURSDAY, 9 JUNE 2022 – AFTERNOON

CHEMISTRY – A2 unit 3

Physical and Inorganic Chemistry

1 hour 45 minutes

	For Examiner's use only		
	Question	**Maximum Mark**	**Mark Awarded**
Section A	**1. to 7.**	**10**	
Section B	**8.**	**14**	
	9.	**20**	
	10.	**12**	
	11.	**24**	
	Total	**80**	

1410U301
01

ADDITIONAL MATERIALS

In addition to this examination paper, you will need a:
- calculator;
- **Data Booklet** supplied by WJEC.

INSTRUCTIONS TO CANDIDATES

Use black ink or black ball-point pen. Do not use gel pen or correction fluid. You may use a pencil for graphs and diagrams only.

Write your name, centre number and candidate number in the spaces at the top of this page.

Section A Answer **all** questions.
Section B Answer **all** questions.

Write your answers in the spaces provided in this booklet. If you run out of space, use the additional page(s) at the back of the booklet, taking care to number the question(s) correctly.

Candidates are advised to allocate their time appropriately between **Section A (10 marks)** and **Section B (70 marks)**.

INFORMATION FOR CANDIDATES

The number of marks is given in brackets at the end of each question or part-question.

The maximum mark for this paper is 80.

Your answers must be relevant and must make full use of the information given to be awarded full marks for a question.

The assessment of the quality of extended response (QER) will take place in **Q.8**(c).